

PERCENTAGE



PERCENTAGES



$$\textcircled{25\%}$$

$$\frac{25}{100}$$

$$22.5\%$$

$$\frac{45}{200}$$

$$\frac{725}{1000}$$

$$\frac{480}{500}$$

$$\checkmark \textcircled{18\%}$$

CONVERSION OF PERCENTAGE TO FRACTION

Percentage to Fraction (Divide by 100)

Eg: 25% = $\frac{25}{100} = \frac{1}{4}$

P F D

60 % =

$$\frac{60}{100} = \frac{3}{5}$$

16 + (2/3) % =

83 + (1/3) % =

$\rightarrow 16 \frac{2}{3} = \frac{50}{3} \% = \frac{\frac{50}{3}}{100} = \frac{50}{3 \times 100} = \frac{1}{6} = \left(\frac{5}{30} \right)$

$83 \frac{1}{3} \% = \frac{250}{3} \% = \frac{\frac{250}{3}}{100} = \frac{250}{3 \times 100} = \frac{1}{12} = \left(\frac{5}{60} \right)$

CONVERSION OF FRACTION TO PERCENTAGE

Fraction to Percentage (Multiply by 100)

$$\textcircled{1} \quad 3/4 = \frac{3}{\cancel{4}} \times \overset{25}{\cancel{100}} = 75\%$$

FPM

$$\textcircled{2} \quad 2/7 = \left(\frac{2}{\cancel{7}} \right) \times 100 = \frac{200}{7}$$

$$\rightarrow 28 \frac{4}{7} \%$$

$$28 \frac{4}{7} \%$$

$$\rightarrow \underline{28.57\%}$$

$$\begin{array}{r} 28.57 \\ 7 \overline{) 200} \\ \underline{-14} \\ 60 \\ \underline{-56} \\ 40 \\ \underline{-35} \\ 50 \\ \underline{-49} \\ 10 \\ \underline{-7} \\ 30 \\ \underline{-28} \\ 20 \\ \underline{-14} \\ 60 \end{array}$$

IMPORTANT CONCEPT

1. A number is increased by 20%

Original Number = 100

New Number = 120

$$\frac{120}{100} = \frac{6}{5}$$

$$\frac{20}{100} = \frac{1}{5}$$

Original Number = 5

New number = $5 + 1 = 6$

$$\begin{array}{r} 5 \times 20 \\ = 10 \\ 5 + 10 \\ = 15 \\ \hline 120 \end{array}$$

IMPORTANT CONCEPT

2. A number is decreased by 25%

$$\text{Orig No} = \underline{100}$$

$$\text{New No.} = 75$$

$$\frac{75}{100} = \frac{\textcircled{3}}{\textcircled{4}}$$

$$25\% = \frac{25}{100} = \frac{1}{4}$$

$$\begin{aligned} \text{Orig No.} &= 4^{\times 25} = 100 \\ \text{New No.} &= 4 - 1 = \textcircled{3}^{\times 25} = 75 \end{aligned}$$

Q. A number is increased by 25%, then it becomes 600. The original number is ____

$$\text{Original no.} = y$$

$$\frac{25}{100} = \frac{1}{4}$$

$$y + 25\% \text{ of } y = 600$$

$$y + \frac{1}{4} \times y = 600$$

$$\frac{5}{4} y = 600$$

$$y = 120 \times 4$$

$$= \underline{480}$$

$$\text{Orig no} = 100 / \checkmark$$

$$\text{New no} = 125\%$$

$$125\% \rightarrow 600$$

$$1\% \rightarrow \frac{600}{125}$$

$$100\% \rightarrow \frac{600}{125} \times 100$$

$$\rightarrow 120 \times 4$$

$$= 480$$

$$25\% = \frac{25}{100} = \frac{1}{4}$$

$$\text{Orig no.} = 4$$

$$\text{New no.} = 4 + 1$$

$$= 5$$

$$5 \rightarrow 600$$

$$1 \rightarrow \frac{600}{5}$$

$$4 \rightarrow \frac{600}{5} \times 4$$

$$\rightarrow 120 \times 4$$

$$= 480$$

Q. A number is decreased by 20%, then it becomes 120. The original number is ____

"Knowledge Gate is One of the Best Platform for Computer Science Aspirants. Sanchit Sir and Yash Sir are Best Teacher for Computer Science Students. Knowledge Gate Courses are very Productive and created a lot of interest in Aptitude Tricks, Programming Fundamentals and Coding. It was just like Boom in Pandemic Lockdown. And During Class Yash Sir always Motivated us and created a lot of Enthusiasm. It was the Best of 2020. Just Because of it I cracked 3 Companies. I am always Proud to be a Student of Knowledge Gate." – **BHANU PRATAP SINGH**, Placed in TCS Ninja, TCS Digital, Hexaware, Accenture,
Prepared from: KG Paid Courses

Q. A number is decreased by 20%, then it becomes 120. The original number is ____

$$20\% = \frac{20}{100} = \frac{1}{5}$$

Orig no = 5

new no = $5 - 1 = 4$

$4 \rightarrow 120$

$1 \rightarrow \frac{120}{4} = 30$

$5 \rightarrow 30 \times 5 = \underline{\underline{150}}$

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Q. A number is decreased by $33\frac{1}{3}\%$, then it becomes 180. The original number is ____

$$33\frac{1}{3}\%$$

accenture

Q. A number is decreased by $33\frac{1}{3}\%$, then it becomes 180. The original number is

0 sec
↓
15 sec

$$33\frac{1}{3}\%$$

$$\rightarrow \frac{1}{3}$$

$$\begin{aligned}\text{Orig no} &= \textcircled{3} \\ \text{New no} &= 3 - 1 = 2\end{aligned}$$

accenture

$$2 \rightarrow 180$$

$$3 \rightarrow 9$$

$$1 \rightarrow 180/2 = 90$$

$$\begin{aligned}3 &\rightarrow 90 \times 3 \\ &= \underline{270}\end{aligned}$$

Q. The total income Yash Sir of 2017, 2018, 2019 is Rs. 364. Every year the salary increases by 20%. What is the income in 2017?



**WHENEVER YOU ASK MONEY
FROM YOUR DAD**

Kya karenge aap itne rupayon ka?

MINUTE
GRAB

Q. The total income Yash Sir of 2017, 2018, 2019 is Rs. 364. Every year the salary increases by 20%. What is the income in 2017? $20\% = \frac{1}{5}$



$$\begin{aligned}
 2017 &\rightarrow 1 \\
 2018 &\rightarrow 1 + \frac{1}{5} = \frac{6}{5} \\
 2019 &\rightarrow \frac{6}{5} + \frac{6}{5} \times \frac{1}{5} \\
 &\rightarrow \frac{6}{5} + \frac{6}{25} \\
 &\rightarrow \frac{30}{25} + \frac{6}{25} = \frac{36}{25}
 \end{aligned}$$

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Q. The total income Yash Sir of 2017, 2018, 2019 is Rs. 364. Every year the salary increases by 20%. What is the income in 2017?



$$\begin{aligned}
 2017 &\rightarrow 1 \\
 2018 &\rightarrow \frac{6}{5} \\
 2019 &\rightarrow \frac{36}{25}
 \end{aligned}$$

19 ✓

$$\frac{1}{1} + \frac{6}{5} + \frac{36}{25}$$

$$\frac{25}{25} + \frac{30}{25} + \frac{36}{25}$$

$$\Rightarrow \frac{25 + 30 + 36}{25}$$

$$\frac{91}{25}$$

$$\frac{91}{25} \rightarrow 364$$

$$\frac{91}{25} \rightarrow 364$$

$$1 \rightarrow 364 \times \frac{25}{91}$$

1 → 100

THREE CONFUSING QUESTIONS

~~####~~

Q1: What % of x is y ?

Q2: y is what % of x ?

Q3: What % is y of x ?

THREE CONFUSING QUESTIONS

Q1: What % of x is y?

Q2: y is what % of x?

Q3: What % is y of x?

$$P\% \text{ of } x = y$$

$$\frac{P}{100} \times x = y$$

$$\frac{Px}{100} = y$$

$$P = \frac{y}{x} \times 100$$

$$\boxed{\frac{y}{x} \times 100}$$

THREE CONFUSING QUESTIONS

Q2: y is what % of x ?

y is p% of x

$$y = p\% \text{ of } x$$

$$y = \frac{p}{100} \times x$$

$$y = \frac{px}{100} \quad \Leftarrow$$

$$p = \frac{y}{x} \times 100$$

$$\frac{y}{x} \times 100$$

THREE CONFUSING QUESTIONS

Q3: What % is y of x ?

KG

THREE CONFUSING QUESTIONS

Q3: What % is y of x ?

p% is y of x

p% of x is y

$$\frac{p}{100} \times x = y$$

$$\frac{p \times x}{100} = y$$

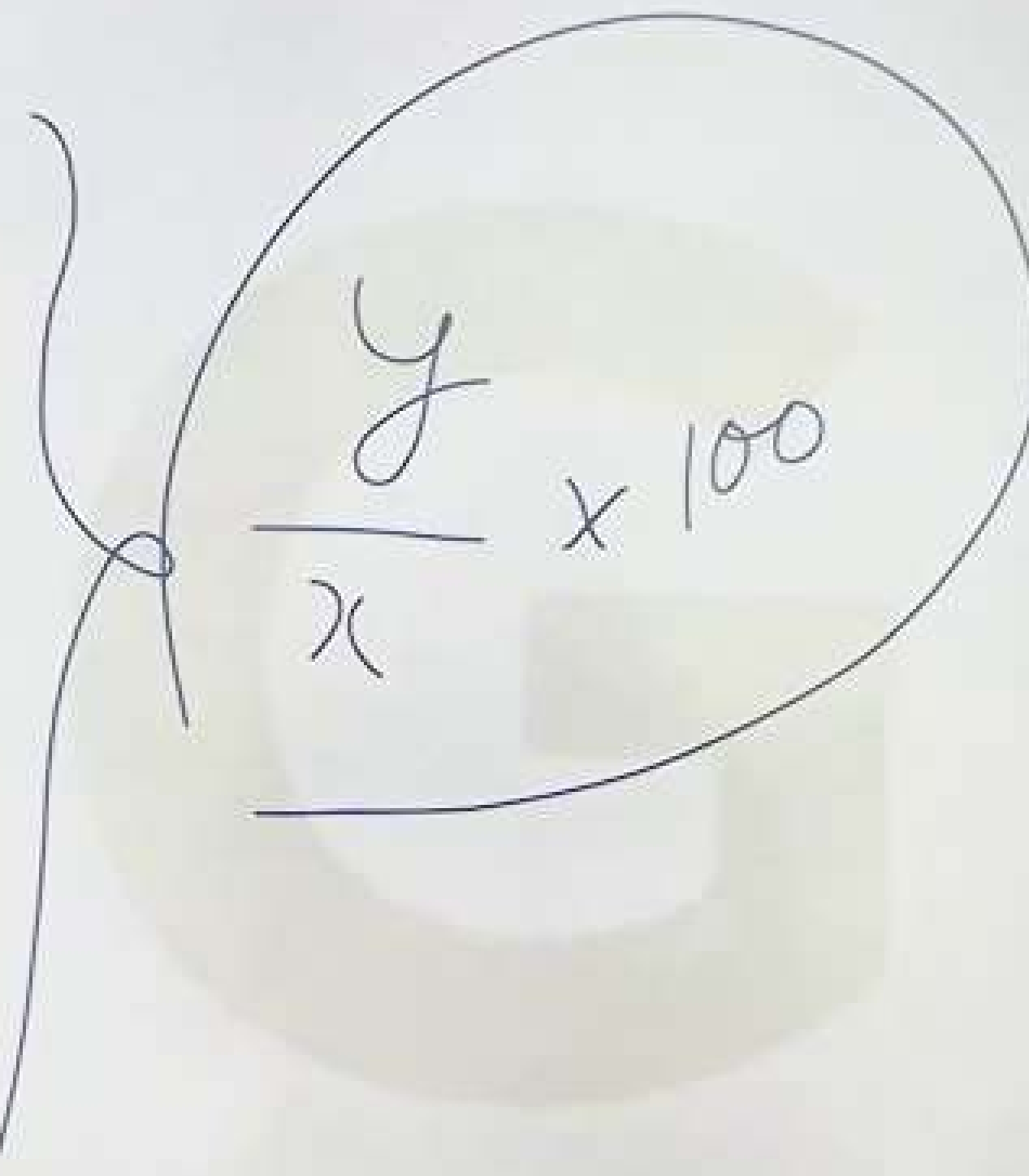
$$p = \frac{y}{x} \times 100$$

THREE CONFUSING QUESTIONS

Q1: What % of x is y ?

Q2: y is what % of x ?

Q3: What % is y of x ?


$$\frac{y}{x} \times 100$$

THREE CONFUSING QUESTIONS

Q1: 32 is what % of 80 ?

Q2: What % of 26 is 52 ?

Q3: 18% of a number is 90, then find the number?

THREE CONFUSING QUESTIONS

Q1: 32 is what % of 80 ? $\frac{32}{80} \times 100 = 40\%$

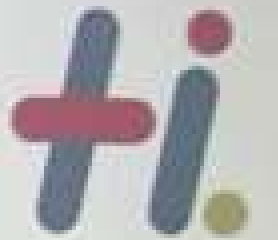
Q2: What % of 26 is 52 ?

$$40\% \text{ of } 80$$
$$\frac{40}{100} \times 80 = 32$$

Q3: 18% of a number is 90, then find the number?

THREE CONFUSING QUESTIONS

Q2: What % of 26 is 52 ?



HEXAWARE

$$\frac{52}{26} \times 100 = 200\%$$

✓
6 months

THREE CONFUSING QUESTIONS

Q3: 18% of a number is 90, then find the number?

$$18\% \text{ of } y = 90$$

$$\frac{18}{100} \times y = 90$$

$$y = \underline{500}$$

Que: If $x/y=4$ and y is not '0' what % of x is $2x-y$

- a) 150%
- b) 175%
- c) 200%
- d) 250%
- e) Cannot be determined
- ☒ Data Insufficient

40 sec

"Yash sir's guidance, motivation always drives me to achieve the goal. His teaching techniques are too good and the whole knowledge gate team is so helpful in terms of doubt solving, problem encounter." – **SOUNAM GHOSH**, Placed in **TCS, Cognizant**, Prepared from: KG Paid Courses

Que: If $x/y=4$ and y is not '0' what % of x is $2x-y$

- a) 150%
 - b) 175% ✓
 - c) 200%
 - d) 250%
 - e) Cannot be determined
- ⊗ Data Insufficient

$$\frac{x}{y} = 4$$

$$\frac{y}{x} = \frac{1}{4}$$

$$\frac{2x-y}{x} \times 100$$

$$\left(\frac{2x}{x} - \frac{y}{x} \right) \times 100$$

$$\left(2 - \frac{1}{4} \right) \times 100$$

$$\frac{7}{4} \times 100 \Rightarrow \underline{175\%}$$

40 sec

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Q. The price of ^{Petrol}~~sugar~~ is increased by 10%, in order to keep the expenditure as constant, by what % consumption must be reduced?



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$$\uparrow \text{Price} \times \text{Consumption} \downarrow = \text{Expenditure}$$

$$100 \times 6 \text{ litre} = 600$$

$$200 \times 3 \text{ litre} = 600$$

$$\uparrow x \times y \downarrow = \text{const}$$



Q. The price of sugar is increased by 10%, in order to keep the expenditure as constant, by what % consumption must be reduced?

201. $\uparrow$ Speed $\times$ time $\downarrow$ = (dist)

$S = \frac{d}{t}$

$x y = \text{const}$

$1000000 \times 5 = 50,00000$

Price $\times$ Sale = revenue



Q. The price of sugar is increased by 10%, in order to keep the + expenditure as constant, by what % consumption must be reduced?

$$\left(\frac{x}{100 + x} \times 100 \right)$$

$$\frac{10}{100 + 10} \times 100$$

$$= \frac{10}{110} \times 100$$

$$= \frac{100}{11} \%$$

$$= 9 \frac{1}{11} \%$$



Q. The price of ^{Onion} ~~tea powder~~ decreased by 25%, in order to keep the expenditure as constant, by what % consumption must be increased?

$$\frac{x}{100+x} \times 100$$

$$\frac{25}{100+25} \times 100$$

$$\frac{25}{100-25} \times 100$$

$$\frac{1}{3} \times 100$$

$$= \frac{100}{3} = 33.33\%$$

Q. A is 25% less than B. By what % is B more than A?

$$\frac{x}{100 \pm x} \times 100$$

$$\frac{25}{100 \pm 25} \times 100$$

$$\Rightarrow \frac{25}{100 - 25} \times 100$$

$$= \frac{25}{75} \times 100$$

$$100/3\% =$$

33.33%

Q. Indian Film Industry is $\frac{2}{7}$ more rich than Indian TV Industry. By what % is Indian TV Industry less rich than Indian Film Industry?

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Indian film industry $\Rightarrow 2$
 TV industry $\Rightarrow 7$
 Film industry $\Rightarrow 7 + 2 = 9$

$$7 + \frac{2}{7} \times 7 = 9$$

$$\frac{200}{9} \% \Rightarrow 22.22\%$$

What On Indian TV serials

1. **5%** Story
2. **20%** Ads
3. **25%** Crying
4. **50%** -

$$\frac{2}{9} \times 100$$

Dhoomtana..Dhoomtana
.dhomm Tere Na Tere
Na



Que: If the price of gold increases by 40%, find by how much the quantity of ornaments must be reduced so that the expenditure may remain the same as before?

- A) 28.57% B) 31.03% C) 36.24% D) 44.52%

accenture

**If Throwing Chappal was
an Olympic Game, Indian
Mothers would Win a
Gold Medal everytime**



“Knowledge Gate course helped me in my placement. With the proper guidance of Yash Sir and the taking mock tests also boost my confidence.” – RAHUL KUMAR,
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Que: If the price of gold increases by 40%, find by how much the quantity of ornaments must be reduced so that the expenditure may remain the same as before?

- A) 28.57% ✓ B) 31.03% C) 36.24% D) 44.52%

$$\frac{x}{100+x} \times 100$$

$$\left(\frac{2}{7} \right) \times 100$$

$$\frac{40}{100+40} \times 100 \Rightarrow \frac{2}{7} \times 100$$

28.57%

accenture

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